TriNetraAI — Subsystem Junction Audit & Resolution Report

Comprehensive Architectural Analysis & Fixes: **Person 1** ↔ **Person 2** and **Person 4** ↔ **Person 3**

Subsystem Junction 1: Person 1 (Ingest) ↔ Person 2 (Semantic)	**Subsystem Junction 2:** Person 4 (QA Filter) ↔ Person 3 (Detective)
Audit Status: All 9 Loopholes Formally Fixed	**Target Branch:** integrated-trinetra-ai

1. Junction Audit: Person 1 (Data Ingestion) ↔ Person 2 (Semantic AI)

• **Loophole 1: Subdirectory Traversal Mismatch:** Person 1 outputs tiles in data/tiles/<scene>/. Person 2 used shallow iterdir(), yielding an empty list [] and FileNotFoundError.

■ **Fix:** Upgraded to recursive rglob('*') with automatic fallback to tiles_metadata.csv registry.

• **Loophole 2: Multi-Band & 16-Bit Radiometric Incompatibility:** Person 1 writes 16-bit GeoTIFFs (4–13 bands). Person 2 used PIL Image.open().convert('RGB'), which crashes on >4 bands and corrupts 16-bit reflectances into pitch-black arrays.

■ **Fix:** Integrated rasterio reader with 2%–98% percentile contrast normalization, matching RemoteCLIP expectations.

• **Loophole 3: Ingestion Contract Bypass:** Person 2 ignored Person 1's SQLite/CSV spatial provenance registry and re-scanned disk manually.

■ **Fix:** Bound embedding generation directly to consume data/tiles_metadata.csv as single source of truth.

• **Loophole 4: Hardcoded Remote Qdrant Host:** Hardcoded http://localhost:6333 crashed on air-gapped field laptops without Docker.

■ **Fix:** Enabled embedded local storage fallback (QdrantClient(path='data/qdrant_local')).

2. Junction Audit: Person 4 (QA False-Alarm Filter) ↔ Person 3 (Temporal Detective)

• **Loophole 1: Fixed SCL Band Assumption (Band 5 Crash):** get_quality_mask_s2 hardcoded scl_band_index=4. On 3-band optical or 4-band Landsat imagery, this triggered IndexError and silently disabled QA filtering.

■ **Fix:** Implemented dynamic band inspection via rasterio. If bands < 5, automatically activates optical brightness/cloud thresholding.

• **Loophole 2: Excessive Cloud Cover Compute Waste & False Semantic Alerts:** When a tile had >85% monsoon cloud cover, Person 3 still ran expensive BIT Transformer inference. Residual eroded cloud slivers were misclassified by RemoteCLIP as 'Water/Flood'.

■ **Fix:** Added **Early-Exit Occlusion Gate** in compare_tiles(): If valid ground < 15%, skips BIT inference immediately and returns skipped_excessive_cloud_cover.

• **Loophole 3: Array Coordinate & Channel Orientation Mismatch:** Person 4 used tifffile while Person 3 used rasterio, risking transposed axis ordering.

■ **Fix:** Standardized both modules onto rasterio for unified CRS and spatial array alignment.

• **Loophole 4: Fixed 2-Pixel Halo Erosion Radius:** A fixed 2-px erosion allowed semi-transparent cirrus halos to bleed past masks on high-resolution tiles.

■ **Fix:** Replaced with adaptive kernel scaling: erode_pixels = max(2, min(H, W) // 128).

• **Loophole 5: Bypassed Sensor-Agnostic Interface Router:** detective_api.py hardcoded direct imports rather than routing through Person 4's sensor adapter.

■ **Fix:** Routed all QA calls through get_quality_mask(sensor_type='auto', ...) supporting Sentinel-2, Landsat, and offline .npy masks.

3. Master Remediation Matrix

Subsystem Junction	Identified Loophole	Severity	Verification Status
Person 1 ↔ Person 2	Subdirectory scan & 16-bit PIL crash	CRITICAL	**RESOLVED & PUSHED**
Person 1 ↔ Person 2	Contract bypass & Qdrant HTTP crash	MEDIUM	**RESOLVED & PUSHED**
Person 4 ↔ Person 3	Fixed SCL index 4 crash on 3/4-band TIFs	CRITICAL	**RESOLVED & PUSHED**
Person 4 ↔ Person 3	Excessive cloud cover compute & false alerts	HIGH	**RESOLVED & PUSHED**
Person 4 ↔ Person 3	Array alignment & adaptive halo erosion	MEDIUM	**RESOLVED & PUSHED**